

FCFS CPU Scheduling

Aim

To write and execute a Python program to implement the First-Come, First-Served (FCFS) CPU scheduling algorithm and calculate the waiting time and turnaround time for each process, along with their averages.

Program

```
GNU nano 8.7.1                                fcfs.py *
def find_wt(process, n, bt, at):
    wt = [0] * n
    service_time = [0] * n
    service_time[0] = at[0]

    for i in range(1, n):
        service_time[i] = service_time[i - 1] + bt[i - 1]
        wt[i] = service_time[i] - at[i]

        if wt[i] < 0:
            wt[i] = 0
            service_time[i] = at[i]

    return wt

def find_TAT(n, bt, wt):
    tat = [0] * n

    for i in range(n):
        tat[i] = bt[i] + wt[i]

    return tat

def find_avg_time(process, n, bt, at):
    proc_info = sorted(
        zip(process, at, bt),
        key=lambda x: x[1]
    )

    process, at, bt = zip(*proc_info)

    wt = find_wt(process, n, bt, at)
    tat = find_TAT(n, bt, wt)
```

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```
total_wt = sum(wt)
total_tat = sum(tat)

print("Execution order:", " -> ".join(map(str, process)))

print("\nProcess\tArrival\tBurst\tWaiting\tTurnaround")

Settings i in range(n):
    print(
        f"{process[i]}\t{at[i]}\t{bt[i]}"
        f"\t{wt[i]}\t{tat[i]}"
    )

print(f"\nAverage Waiting Time = {total_wt / n:.2f}")
print(f"Average Turnaround Time = {total_tat / n:.2f}")

if __name__ == "__main__":
    process = [1, 2, 3, 4]
    arrival_time = [0, 1, 2, 3]
    burst_time = [5, 3, 8, 6]

    find_avg_time(
        process,
        len(process),
        burst_time,
        arrival_time
    )
```

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Output :

Output for the given process, arrival-time, and burst-time values:

```
python3: can't open file '/home/aayush/fcfs.py': [Errno 2] No such file or directory
aayush@ubuntu:~$ nano fcfs.py
aayush@ubuntu:~$ python3 fcfs.py
Execution order: 1 -> 2 -> 3 -> 4

Process Arrival Burst Waiting Turnaround
1 0 5 0 5
2 1 3 4 7
3 2 8 6 14
4 3 6 13 19

Average Waiting Time = 5.75
Average Turnaround Time = 11.25
aayush@ubuntu:~$
```

Result

The FCFS scheduling program was executed. The average waiting time is 5.75 time units and the average turnaround time is 11.25 time units.